A Report

on

Smart Attendance System using Image Verification and Metadata Validation

carried out as part of the course Project Based Learning- 2/4

Submitted by

Saksham Jain

2427030737

IVSemester

in partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering



MANIPAL UNIVERSITY
JAIPUR

Department of Computer Science & Engineering,
School of Computer Science and Engineering,
Manipal University Jaipur,
Jan-May 2026

A Report

on

Smart Attendance System using Image Verification and Metadata Validation

carried out as part of the course Project Based Learning-2/4

Submitted by

Saksham Jain

2427030737

IVSemester

in partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering

Under the Guidance of :
Guide Name : Dr. Gireesh Kumar Kaushik

Guide Signature(with date) : 11/04/2026

Acknowledgement

This project would not have been completed without the help, support, comments, advice, cooperation and coordination of various people. However, it is impossible to thank everyone individually; I am hereby making a humble effort to thank some of them.

I acknowledge and express my deepest sense of gratitude to my internal supervisor Dr. Gireesh Kumar Kaushik for his/her constant support, guidance, and continuous engagement. I highly appreciate his technical comments, suggestions, and criticism during the progress of this project titles “Smart Attendance System using Image Verification and Metadata Validation”.

I owe my profound gratitude to ***Dr. Neha Chaudhary*** , Head, Department of CSE, for her valuable guidance and for facilitating me during my work. I am also very grateful to all the faculty members and staff for their precious support and cooperation during the development of this project.

Finally, I extend my heartfelt appreciation to my classmates for their help and encouragement.

Saksham Jain

2427030737

MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

Department of Computer Science and Engineering

School of Computer Science and Engineering

Date: 11.04.2026

CERTIFICATE

This is to certify that the project entitled "Smart Attendance System using Image Verification and Metadata Validation" is a bonafide work carried out as PBL-2 ***End Term Assessment*** in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, by ***(Saksham Jain)*** bearing registration number (**2427030737**), during the academic semester IV *of year 2025-2026.*

Place: Manipal University Jaipur, Jaipur

Name of the project guide: Dr. Gireesh Kumar Kaushik

Signature of the project guide: _____

Smart Attendance System using Image Verification and Metadata Validation

This project is a web-based smart attendance solution that allows a user to upload an image for attendance marking. The system verifies the uploaded image using perceptual hashing and metadata validation before marking attendance. It is designed to prevent duplicate submissions, reuse of old images, and proxy attendance attempts. The final prototype uses a frontend interface for image upload and a Flask backend for processing and verification.

TABLE OF CONTENTS

1. Introduction
2. Objective of the Project
3. Brief Description of the Project
4. Technology Used
5. Hardware Requirements
6. Software Requirements
7. Organization Profile
8. Design Description
9. Flow Chart
10. Data Flow Diagrams (DFDs)
11. Entity Relationship Diagram (ER Diagram)
12. Project Description
13. Database Design
14. Table Description
15. File / Database Design
16. Input / Output Form Design
17. Testing & Tools Used
18. Implementation & Maintenance
19. Conclusion and Future Work
20. Outcome
21. Bibliography

1. INTRODUCTION

Attendance management is an important part of academic institutions, training organizations, and workplaces. It is used to maintain discipline, monitor participation, and create official records of presence. Traditional attendance methods such as manual registers, roll calls, fingerprint scanners, RFID cards, and biometric machines are commonly used, but each of these methods has limitations. Manual systems are time-consuming and can be manipulated through proxy attendance. Hardware-based systems require additional devices and maintenance, which increases cost and reduces flexibility.

With the growth of digital learning, online classes, hybrid work environments, and mobile-based systems, there is a growing need for a more flexible attendance solution. A software-based smart attendance system can solve many of the limitations of traditional methods by allowing users to mark attendance through a web interface. This approach is especially useful in situations where physical contact, specialized devices, or in-person verification are not practical.

This project introduces a smart attendance system based on image verification and metadata validation. In this system, a user uploads an image along with a unique user ID. The backend then processes the image, generates a perceptual hash, checks whether the image is similar to a previously uploaded image, and also verifies the image metadata if available. If the uploaded image passes the validation checks, the attendance is marked; otherwise, the image is rejected. This provides a more secure and controlled method of attendance verification.

The major idea behind this project is not only to store attendance but also to ensure that the image submitted is genuine, recent, and not reused. This helps prevent duplicate attendance marking and makes the system more reliable for future extension into face recognition or live camera-based attendance.

2. OBJECTIVE OF THE PROJECT

The main objective of this project is to design and develop a smart attendance system that verifies the authenticity of uploaded images before marking attendance. The system aims to reduce the dependency on manual registers and hardware devices while improving attendance security and efficiency.

The specific objectives of the project are:

- To develop a web-based attendance marking system.
- To allow users to upload images for attendance verification.
- To detect duplicate or similar images using perceptual hashing.
- To validate image metadata such as capture date whenever available.
- To prevent repeated attendance submissions on the same day.
- To store attendance records in a secure database.
- To create a low-cost and easily deployable attendance prototype.
- To provide a foundation for future enhancement using face recognition and live webcam capture.

This project is designed as a practical prototype that can be extended into a full smart attendance system in later stages.

3. BRIEF DESCRIPTION OF THE PROJECT

The proposed system is a browser-based attendance application where users enter their user ID and upload an image. The backend processes the image through multiple validation steps. First, a perceptual hash is created from the uploaded image. This hash is compared with

previously stored hashes to identify if the image is too similar to an older one. Then the system checks the image metadata, particularly the capture date, if such metadata is available. Finally, the system checks whether attendance has already been marked for that user on the same day.

If all checks pass, the attendance is marked and stored in the database. If the system detects a duplicate image, a reused image, or repeated attendance for the day, it rejects the submission. The system is built using a simple frontend and a Flask backend, making it suitable for a small-scale project prototype, classroom demonstration, or future expansion.

The project focuses on image-based verification rather than fully automated face recognition. This makes the first version lightweight and easy to implement, while still demonstrating intelligent validation logic.

4. TECHNOLOGY USED

The project uses a combination of frontend, backend, and database technologies.

Frontend

The frontend is developed using a web interface that accepts:

- User ID
- Image upload
- Submit button
- Response message display

In the working prototype, the frontend was implemented using **React with Vite**, though the same can also be done using plain HTML, CSS, and JavaScript.

Backend

The backend is developed using:

- **Python**
- **Flask**
- **Flask-CORS**

Flask handles the HTTP requests and response processing, while Flask-CORS allows the frontend and backend to communicate across different ports during development.

Image Processing

The following libraries are used:

- **Pillow** for image opening and metadata extraction
- **ImageHash** for perceptual hashing and similarity detection

Database

The project uses:

- **SQLite**, a lightweight embedded relational database

SQLite stores:

- User ID
- Image hash
- Last attendance date

This technology stack was chosen because it is simple, lightweight, open-source, and appropriate for a college-level project.

5. HARDWARE REQUIREMENTS

The project does not require specialized hardware. It can run on a regular laptop or desktop computer.

Minimum Hardware Requirements:

- Processor: Dual-core or better
- RAM: 4 GB or above
- Storage: At least 1 GB free space
- Camera: Optional, for future live image capture
- Display: Standard monitor or laptop screen
- Internet: Optional for local development, not mandatory for offline testing

Recommended Hardware:

- Processor: Intel i5 or equivalent
- RAM: 8 GB or above
- SSD storage for faster performance

Since the current prototype uses image upload, no additional biometric or RFID device is required.

6. SOFTWARE REQUIREMENTS

The software requirements for the project are:

- Operating System: macOS / Windows / Linux
- Python 3.x

- VS Code or any Python IDE
- Flask
- Flask-CORS
- Pillow
- ImageHash
- SQLite
- Node.js and npm if frontend is developed with Vite/React
- Web browser such as Chrome or Safari

The system can be run locally without requiring any paid software or cloud deployment.

7. ORGANIZATION PROFILE

If applicable, this section may describe the institute, department, or organization where the project is being developed.

The project has been developed as part of the DT&I project work in the Department of Computer Science and Engineering. The objective of this academic project is to encourage design thinking, problem solving, and implementation of a practical digital solution. The project is intended to demonstrate both technical ability and the ability to identify a real-world problem and solve it using software tools.

8. DESIGN DESCRIPTION

The design of the system is divided into three main components:

Frontend Design

The frontend provides a simple user interface that accepts:

- User ID input
- Image file selection
- Submit action
- Result display

The interface is kept minimal so that the system is easy to use and easy to test.

Backend Design

The backend processes the uploaded image and performs the following tasks:

- Accepts the request from frontend
- Reads the image

- Generates perceptual hash
- Extracts EXIF metadata
- Checks duplicate attendance
- Validates the record
- Sends back response

Database Design

The database stores:

- user ID
- image hash
- attendance date

This ensures that old records can be used for comparison and the attendance history is preserved.

The overall design is modular, making it easy to modify or extend in the future.

9. FLOW CHART

The flow of the system is as follows:

1. Start
2. Enter User ID
3. Upload Image
4. Generate perceptual hash
5. Compare with stored hash
6. Extract metadata
7. Check if attendance already marked
8. If valid, save attendance
9. If invalid, reject submission
10. End

Flowchart Description:

The user first enters the required details and uploads an image. The system processes the image and performs validation checks. If the image is unique, recent, and the user has not already marked attendance for the day, the attendance is stored successfully. Otherwise, the submission is rejected.

10. DATA FLOW DIAGRAMS (DFDs)

Level 0 DFD

At the highest level, the system can be represented as:

- User
- Smart Attendance System
- Database

The user sends the image and user ID to the system. The system validates the data and stores the result in the database.

Level 1 DFD

The Level 1 diagram can include the following processes:

- Image upload
- Image verification
- Metadata extraction
- Duplicate checking
- Attendance marking
- Data storage

Data Flow:

User input → Frontend → Flask backend → Validation module → SQLite database → Response to user

11. ENTITY RELATIONSHIP DIAGRAM (ER DIAGRAM)

The main entities in the system are:

User

Attributes:

- User ID
- Name (optional)

Attendance

Attributes:

- Attendance ID
- User ID
- Image hash
- Date of attendance

- Status

Relationship:

A user can have multiple attendance records over time, but only one record can be created per day.

The ER diagram shows how the user entity is linked with attendance records through the user ID.

12. PROJECT DESCRIPTION

The project works as a smart attendance verification system. The user interacts with the frontend, enters a unique ID, uploads an image, and submits the request. The backend receives the image and performs verification using a series of image processing and database checks.

The first step is image hash generation. The uploaded image is converted into a perceptual hash that represents the visual structure of the image. This hash is used to compare the current image with earlier uploaded images. If the hash difference is too small, it means the image is very similar to a previous image and may be a duplicate.

The system then attempts to extract metadata from the image, especially the capture date. If metadata is present, it is compared with the current date. This helps identify outdated or reused images. Finally, the system checks whether attendance has already been marked for that user on the same day.

If the image is unique and valid, the attendance record is stored in the database and a success message is sent to the user. If not, the system returns a rejection message. This ensures that the system is secure and resistant to basic forms of misuse.

13. DATABASE

The project uses SQLite as the database engine. SQLite was selected because it is lightweight, easy to use, and suitable for small-scale local applications.

The database stores attendance data in a table named `users` or a similar attendance table depending on implementation. The stored information generally includes:

- User ID
- Image hash
- Last attendance date

The database allows the system to:

- Verify whether the user has already marked attendance
- Compare current image with previous images
- Keep a record of attendance submissions

14. TABLE DESCRIPTION

A sample table structure may be described as follows:

Field Name	Data Type	Description
user_id	TEXT	Unique identifier of the user
image_hash	TEXT	Perceptual hash of uploaded image
last_attendance	TEXT / DATE	Date when attendance was last marked

If additional tables are used, such as a separate attendance history table, they can be described in the same format.

The table is designed in a simple way so that lookups and updates can be performed efficiently.

15. FILE / DATABASE DESIGN

The project contains multiple source files, each serving a specific role:

Backend Files

- `app.py` — main Flask application and route handling
- `database.py` — database connection and user record functions
- `utils.py` — helper functions for hashing and metadata extraction
- `requirements.txt` — list of Python dependencies

Frontend Files

- `index.html` / React root file
- `App.jsx` or equivalent component
- `main.jsx`
- `index.css`

Database File

- `attendance.db` — SQLite database file created automatically

Uploads Folder

- `uploads/` — stores uploaded image files locally

This modular design improves readability and makes debugging easier.

16. INPUT / OUTPUT FORM DESIGN

Input Form

The input form contains:

- User ID field
- Image selection field
- Submit button

The form is designed to be minimal and user-friendly.

Output Form

The output is displayed as a text response on the interface, such as:

- Attendance marked successfully
- Image too similar
- Metadata mismatch
- Attendance already marked today
- Missing data

This immediate feedback helps the user understand whether the submission succeeded or failed.

17. TESTING & TOOLS USED

Testing

The project was tested using multiple test scenarios:

- Uploading a valid new image
- Uploading the same image again
- Uploading a different image for the same user
- Uploading an image from a previous date
- Attempting multiple submissions on the same day

The testing confirmed that:

- Attendance is marked successfully for valid input
- Duplicate submissions are rejected
- Same-day repeated marking is blocked
- Metadata checks work when data is available

Tools Used

- VS Code
- Python terminal
- Browser developer tools
- Flask local server
- SQLite local database

These tools were sufficient to build, test, and debug the application.

18. IMPLEMENTATION & MAINTENANCE

The project was implemented in a step-by-step manner. First, the backend was created and tested. Then the frontend was developed and connected to the backend. After that, image hashing and metadata validation were added. Finally, database storage and duplicate check logic were implemented.

Maintenance of the project is simple because the architecture is modular. If needed, the system can be updated by:

- improving similarity threshold
- adding face recognition
- adding webcam capture
- improving UI design
- changing database structure
- deploying the project online

Because the code is separated into modules, maintenance and future enhancement can be done with minimal disruption.

19. CONCLUSION AND FUTURE WORK

The project successfully demonstrates a smart attendance system that uses image verification and metadata validation. It provides a practical alternative to manual and hardware-based attendance methods. The system is able to detect duplicate images, validate image freshness, and prevent repeated attendance entries on the same day.

The current prototype shows that image-based attendance can be implemented effectively using simple and lightweight technologies. The project also proves that perceptual hashing and metadata checking can be combined to improve attendance security.

Future Work:

- Face recognition integration
- Live webcam attendance capture
- Anti-spoofing and liveness detection
- Online deployment
- Attendance dashboard
- Excel/PDF export of attendance records
- Mobile support

These enhancements can turn the prototype into a full smart attendance solution.

20. OUTCOME

The project outcome can be described in 3 lines as requested:

A functional smart attendance prototype has been developed successfully using image verification and metadata validation. The system can detect duplicate submissions and restrict repeated attendance marking. The project lays a strong foundation for future development into a face recognition and live camera-based attendance system.

21. BIBLIOGRAPHY

Use this format in your report:

[1] Flask Documentation. Python Web Framework.
[2] Pillow Documentation. Python Imaging Library for image processing.
[3] ImageHash Documentation. Library for perceptual hashing.
[4] SQLite Documentation. Lightweight embedded database documentation.
[5] Research articles on image-based attendance systems and digital image forensics.